

Dynamic Bipartite Network: Mathematical Derivations

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1 Setup

Let $G_t = (V_{1,t}, V_{2,t}, E_t)$ represent a dynamic bipartite graph observed at time t , where V_1 and V_2 denote two disjoint families of nodes and E represents the set of undirected edges between two nodes of different families. Suppose that (at t) family 1 has $N_{1,t} = |V_{1,t}|$ nodes whereas the number of nodes in family 2 is $N_{2,t} = |V_{2,t}|$. For a pair of nodes p, q , let $z_{pq,t} \in \{1, \dots, K_1\}$ denote the latent group that node $p \in V_{1,t}$ of family 1 instantiates when interacting with node $q \in V_{2,t}$ whose latent group membership is denoted by $u_{pqt} \in \{1, \dots, K_2\}$. Further, let $y_{pqt} = 1$ if there exists a directed edge from node p to q for $(p, q) \in E_t$, and $y_{pqt} = 0$ otherwise.

Assume further that the network at time t is in one of M latent states, and that a Markov process governs transitions from one state to the next. We then assume the mixed-membership vectors are generated according to Markov-dependent mixtures with Dirichlet distributions whose concentration parameters are functions of node covariates:

$$\begin{aligned}\boldsymbol{\pi}_{pt} \mid \boldsymbol{\beta}_1 &\sim \sum_{m=1}^M \mathbb{P}(S_t = m \mid S_{t-1}) \times \text{Dirichlet} \left(\left\{ \exp(\mathbf{x}_{pt}^\top \boldsymbol{\beta}_{1gm}) \right\}_{g=1}^{K_1} \right) \\ \boldsymbol{\psi}_{qt} \mid \boldsymbol{\beta}_2 &\sim \sum_{m=1}^M \mathbb{P}(S_t = m \mid S_{t-1}) \times \text{Dirichlet} \left(\left\{ \exp(\mathbf{x}_{qt}^\top \boldsymbol{\beta}_{2hm}) \right\}_{h=1}^{K_2} \right)\end{aligned}$$

, where the vector of predictors $\mathbf{x}_{it}$ is allowed to vary over time, vectors $\boldsymbol{\beta}_{1gm}$ and $\boldsymbol{\beta}_{2hm}$, indexed by state m in the Markov process, contain regression coefficients associated with the g th and h th groups of vertex families 1 and 2. Further, the random states are generated according to:

$$S_t \mid S_{t-1} \sim \text{Categorical}(\mathbf{A}_n)$$

, where $\mathbf{A}$ is the transition matrix. We define a uniform prior over the initial state S_1 and independent symmetric Dirichlet prior distribution for the rows of $\mathbf{A}$. Then, as common in mixed-membership SBMs, we define a categorical sampling model for the dyad-state specific group memberships, $z_{pq,t}$ and $u_{pq,t}$ as:

$$z_{pq,t} \mid \boldsymbol{\pi}_{pt} \sim \text{Categorical}(\boldsymbol{\pi}_{pt}), \quad u_{pq,t} \mid \boldsymbol{\psi}_{qt} \sim \text{Categorical}(\boldsymbol{\psi}_{qt})$$

The model is then completed by defining a $K_1 \times K_2$ blockmodel $\mathbf{B}$, with its B_{gh} element giving the log odds of forming an edge between any two latent group members. Therefore:

$$y_{pqt} \mid z_{pqt}, u_{pqt}, \mathbf{B}, \boldsymbol{\gamma} \stackrel{\text{indep.}}{\sim} \text{Bernoulli} \left(\text{logit}^{-1}(B_{z_{pqt}, u_{pqt}} + \mathbf{d}_{pqt}^\top \boldsymbol{\gamma}) \right)$$

In sum, the data-generating process can be described as follows:

1. For each time period $t > 1$, draw a historical state $S_t \mid S_{t-1} = n \sim \text{Categorical}(\mathbf{A}_n)$
2. For each node $p \in V_1$ and $q \in V_2$ at time t , draw state-dependent mixed-membership vectors $\boldsymbol{\pi}_{pt} \mid \boldsymbol{\beta}_1, S_t = m \sim \text{Dirichlet} \left(\left\{ \exp(\mathbf{x}_{pt}^\top \boldsymbol{\beta}_{1gm}) \right\}_{g=1}^{K_1} \right)$ and $\boldsymbol{\psi}_{qt} \mid \boldsymbol{\beta}_2, S_t = m \sim \text{Dirichlet} \left(\left\{ \exp(\mathbf{x}_{qt}^\top \boldsymbol{\beta}_{2hm}) \right\}_{h=1}^{K_2} \right)$

3. For each pair of nodes $(p, q) \in E_t$ at time t ,

- Sample a group indicator $z_{pq,t} \mid \boldsymbol{\pi}_{pt} \sim \text{Categorical}(\boldsymbol{\pi}_{pt})$
- Sample a group indicator $u_{pq,t} \mid \boldsymbol{\psi}_{qt} \sim \text{Categorical}(\boldsymbol{\psi}_{qt})$
- Sample a link between them $y_{pqt} \mid z_{pqt}, u_{pqt}, \mathbf{B}, \boldsymbol{\gamma} \stackrel{\text{indep.}}{\sim} \text{Bernoulli}(\text{logit}^{-1}(B_{z_{pqt}, u_{pqt}} + \mathbf{d}_{pqt}^\top \boldsymbol{\gamma}))$

Therefore, the DGP gives the full joint distribution of data and latent variables in the model given a set of global hyper-parameters $(\boldsymbol{\beta}, \boldsymbol{\gamma}, \mathbf{B})$ and covariates $(\mathbf{D}, \mathbf{X})$ as:

$$\begin{aligned}
f(\mathbf{Y}, \mathbf{Z}, \mathbf{U}, \boldsymbol{\Pi}, \boldsymbol{\Psi}, \mathbf{A} \mid \mathbf{B}, \boldsymbol{\beta}, \boldsymbol{\gamma}) &= P(S_1) \left[\prod_{t=2}^T P(S_t \mid S_{t-1}, \mathbf{A}) \right] \prod_{m=1}^M P(\mathbf{A}_m) \\
&\times \prod_{t=1}^T \prod_{p \in V_{1t}} f(\boldsymbol{\pi}_{pt} \mid \mathbf{X}_1, \boldsymbol{\beta}_1, S_t) \prod_{q \in V_{2t}} f(\boldsymbol{\psi}_{qt} \mid \mathbf{X}_2, \boldsymbol{\beta}_2, S_t) \\
&\times \prod_{t=1}^T \prod_{p, q \in V_{1t} \times V_{2t}} f(y_{pqt} \mid z_{pqt}, u_{pqt}, \mathbf{B}, \mathbf{D}, \boldsymbol{\gamma}) f(z_{pqt} \mid \boldsymbol{\pi}_{pt}) f(u_{pqt} \mid \boldsymbol{\psi}_{qt})
\end{aligned} \tag{1}$$

2 Marginalization

2.1 Marginalizing $\boldsymbol{\Pi}$

Collect and integrate all terms that contain $\boldsymbol{\pi}$:

$$\int \cdots \int \prod_{t=1}^T \prod_{p \in V_{1t}} \left[P(\boldsymbol{\pi}_{pt} \mid \mathbf{X}, \boldsymbol{\beta}_1, S_t) \right] \prod_{qt \in V_{2t}} P(\mathbf{z}_{pqt} \mid \boldsymbol{\pi}_{pt}) d\boldsymbol{\pi}_{1t} \cdots d\boldsymbol{\pi}_{N_{1t}}$$

Denote $\alpha_{ptgm} = \exp(\mathbf{x}_{pt}^\top \boldsymbol{\beta}_{1gm})$, and $\xi_{ptm} = \sum_{g=1}^{K_1} \alpha_{ptgm}$, then $\boldsymbol{\pi}_{pt} \mid \xi_{ptm} \sim \text{Dir}(\xi_{ptm})$. Therefore, plugging in the PDF for Dirichlet yields:

$$\prod_{t=1}^T \prod_{p \in V_{1t}} \int \prod_{m=1}^M \left[\frac{\Gamma(\xi_{ptm})}{\prod_{g=1}^{K_1} \Gamma(\alpha_{ptgm})} \prod_{g=1}^{K_1} \pi_{ptg}^{\alpha_{ptgm}-1} \right]^{s_{tm}} \prod_{q \in V_{2t}} \prod_{g=1}^{K_1} \pi_{ptg}^{z_{pqt,g}} d\boldsymbol{\pi}_{pt}$$

, where $z_{pqt,g} = \mathbb{I}(z_{pqt} = g)$. Define $C_{ptg} = \sum_{q \in V_{2t}} z_{pqt,g}$, applying the trick that for indicator function $s_{tm} = \mathbb{I}(S_t = m)$, $\sum_m s_{tm} x = \prod_m x^{s_{tm}}$ and taking the constant terms out of the integral:

$$\prod_{t=1}^T \prod_{p \in V_{1t}} \prod_{m=1}^M \left[\frac{\Gamma(\xi_{ptm})}{\prod_{g=1}^{K_1} \Gamma(\alpha_{ptgm})} \right]^{s_{tm}} \int \prod_{g=1}^{K_1} \pi_{ptg}^{\sum_{m=1}^M s_{tm} \alpha_{ptgm} + C_{ptg} - 1} d\boldsymbol{\pi}_{pt}$$

The integrand can be recognized as the kernel of a Dirichlet distribution. As the integral is over the entire support of this Dirichlet and must integrate to one, we can compute it as the inverse of the corresponding normalizing constant:

$$\prod_{t=1}^T \prod_{p \in V_{1t}} \prod_{m=1}^M \left[\frac{\Gamma(\xi_{ptm})}{\prod_{g=1}^{K_1} \Gamma(\alpha_{ptgm})} \right]^{s_{tm}} \frac{\prod_{g=1}^{K_1} \Gamma\left(\sum_{m=1}^M s_{tm} \alpha_{ptgm} + C_{ptg}\right)}{\Gamma\left(\sum_{m=1}^M s_{tm} \xi_{ptm} + N_{2t}\right)}$$

, where N_{2t} is the number of nodes in family 2 at time t . Apply the indicator trick again and rearranging the factorials:

$$\prod_{t=1}^T \prod_{p \in V_{1t}} \prod_{m=1}^M \left[\frac{\Gamma(\xi_{ptm})}{\Gamma(\xi_{ptm} + N_{2t})} \prod_{g=1}^{K_1} \frac{\Gamma(\alpha_{ptgm} + C_{ptg})}{\Gamma(\alpha_{ptgm})} \right]^{s_{tm}} \tag{2}$$

2.2 Marginalizing Ψ

Collect and integrate all terms that contain ψ :

$$\int \cdots \int \prod_{t=1}^T \prod_{q \in V_{2t}} \left[P(\psi_{qt} \mid \mathbf{X}, \beta_2, S_t) \right] \prod_{pt \in V_{1t}} P(\mathbf{u}_{pqt} \mid \psi_{qt}) d\psi_{1t} \cdots d\psi_{N_{2t}}$$

Following a similar strategy as 2.1 yields:

$$\prod_{t=1}^T \prod_{q \in V_{2t}} \prod_{m=1}^M \left[\frac{\Gamma(\xi_{qtm})}{\Gamma(\xi_{qtm} + N_{1t})} \prod_{h=1}^{K_2} \frac{\Gamma(\alpha_{qthm} + C_{qth})}{\Gamma(\alpha_{qthm})} \right]^{s_{tm}} \quad (3)$$

, where $C_{qth} = \sum_{p \in V_{1t}} \mathbb{I}(u_{qp,t} = h)$, $\alpha_{qthm} = \exp(\mathbf{x}_{qt}^\top \beta_{2hm})$, and $\xi_{qtm} = \sum_{h=1}^{K_2} \alpha_{qhm}$.

2.3 Marginalizing $\mathbf{A}$

Since the transition probabilities have independent Dirichlet priors, and they are conjugate to the multinomial distribution over states at any given time, we can follow a similar strategy when collapsing the rows of $\mathbf{A}$. More specifically, and focusing on the portion of the joint distribution that involves $\mathbf{A}$, we have

$$\begin{aligned} \int \cdots \int \prod_{t=2}^T P(s_t \mid s_{t-1}, \mathbf{A}) \prod_m P(\mathbf{A}_m) d\mathbf{A}_1 \cdots d\mathbf{A}_M = \\ \int \cdots \int \prod_{t=2}^T \prod_m \prod_n A_{m,n}^{s_{t,n} \times s_{t-1,m}} \prod_m \frac{\Gamma(M\eta)}{\prod_n \Gamma(\eta)} \prod_n A_{m,n}^{\eta-1} d\mathbf{A}_1 \cdots d\mathbf{A}_M \quad (4) \\ = \prod_m \frac{\Gamma(M\eta)}{\Gamma(M\eta + U_{m\cdot})} \prod_n \frac{\Gamma(\eta + U_{m,n})}{\Gamma(\eta)} \end{aligned}$$

, where $U_{m,n} = \sum_{t=2}^T s_{t,n} s_{t-1,m}$ is the number of times the Markov chain transitions from state m to state n , and $U_{m\cdot} = \sum_{t=2}^T \sum_n s_{t,n} s_{t-1,m}$ is the total number of times the Markov chain transitions from m (potentially to stay at m). η is the hyperprior concentration parameter of a symmetric Dirichlet distribution.

2.4 Marginalized Joint Distribution

Plugging Equations (2) to (4) back into Equation (1), we can get the joint distribution collapsed over the mixed-membership vectors and the transition matrix.

$$\begin{aligned} f(\mathbf{Y}, \mathbf{Z}, \mathbf{U} \mid \mathbf{B}, \beta, \gamma) &= \iiint f(\mathbf{Y}, \mathbf{Z}, \mathbf{U}, \Pi, \Psi, \mid \mathbf{B}, \beta, \gamma) d\Pi d\Psi d\mathbf{A} \\ &= P(s_1) \left[\prod_{m=1}^M \frac{\Gamma(M\eta)}{\Gamma(M\eta + U_{m\cdot})} \prod_{n=1}^M \frac{\Gamma(\eta + U_{m,n})}{\Gamma(\eta)} \right] \\ &\times \prod_{t=2}^T \prod_{m=1}^M \prod_{p \in V_{1t}} \left[\frac{\Gamma(\xi_{ptm})}{\Gamma(\xi_{ptm} + N_{2t})} \prod_{g=1}^{K_1} \frac{\Gamma(\alpha_{ptgm} + C_{ptg})}{\Gamma(\alpha_{ptgm})} \right]^{s_{tm}} \\ &\times \prod_{q \in V_{2t}} \left[\frac{\Gamma(\xi_{qtm})}{\Gamma(\xi_{qtm} + N_{1t})} \prod_{h=1}^{K_2} \frac{\Gamma(\alpha_{qthm} + C_{qth})}{\Gamma(\alpha_{qthm})} \right]^{s_{tm}} \\ &\times \prod_{p, q \in V_{1t} \times V_{2t}} \left[\prod_{g=1}^{K_1} \prod_{h=1}^{K_2} \left(\theta_{pqt, z_{pqt}, u_{pqt}}^{y_{pqt}} (1 - \theta_{pqt, z_{pqt}, u_{pqt}})^{1-y_{pqt}} \right)^{z_{pqt, g} \times u_{pqt, h}} \right] \end{aligned} \quad (5)$$

, where $\theta_{pqt, z_{pqt}, u_{qpt}} = \text{logit}^{-1}(B_{z_{pqt}, u_{qpt}} + \mathbf{d}_{pq}^\top \boldsymbol{\gamma})$ is the probability of a tie formation between p in family 1 and q in family 2 at time t

3 Estimation via Variational EM

Define a factorized distribution over the latent variables $\mathbf{L} := \{\mathbf{Z}, \mathbf{U}, \mathbf{S}\}$:

$$\tilde{Q}(\mathbf{L} \mid \boldsymbol{\Phi}, \boldsymbol{\Lambda}, \boldsymbol{\Delta}) = \prod_{t=1}^T Q_1(\mathbf{s}_t \mid \boldsymbol{\phi}_t) \prod_{p, q \in V_{1t} \times V_{2t}} Q_2(\mathbf{z}_{pq, t} \mid \boldsymbol{\lambda}_{pq, t}) Q_2(\mathbf{u}_{qp, t} \mid \boldsymbol{\delta}_{qp, t}) \quad (6)$$

, where $\boldsymbol{\phi}_t$, $\boldsymbol{\lambda}_{pq, t}$, and $\boldsymbol{\delta}_{qp, t}$ are variational parameters. We can then find the lower bound for the log marginal probability of the network data $\mathbf{Y}$ by applying Jensen's inequality:

$$P(\mathbf{Y} \mid \boldsymbol{\beta}, \boldsymbol{\gamma}, \mathbf{B}) \geq \mathcal{L} := \mathbb{E}_{\tilde{Q}} [\log P(\mathbf{Y}, \mathbf{L} \mid \boldsymbol{\beta}, \boldsymbol{\gamma}, \mathbf{B})] - \mathbb{E}_{\tilde{Q}} [\log \tilde{Q}(\mathbf{L} \mid \boldsymbol{\Phi}, \boldsymbol{\Lambda}, \boldsymbol{\Delta})] \quad (7)$$

To approximate the true posterior over the latent variables, we optimize this lower bound by iterating between finding an optimal $\tilde{Q}$ (the E-step) and optimizing the corresponding lower bound with respect to the hyper-parameters $\mathbf{B}, \boldsymbol{\beta}, \boldsymbol{\gamma}$ (the M-step).

3.1 The E-Steps

3.1.1 E step 1: $\mathbf{Z}$ and $\mathbf{U}$

Variational parameters λ_{pqt} and δ_{pqt} are updated by restricting eq. (5) to the terms that only contain $\mathbf{z}_{pqt}$ and $\mathbf{u}_{pqt}$ and taking the logarithm of the resulting expression. First, consider $\mathbf{z}_{pqt}$:

$$\begin{aligned} \log P(\mathbf{Y}, \mathbf{L} \mid \mathbf{B}, \boldsymbol{\beta}_1, \boldsymbol{\beta}_2, \boldsymbol{\gamma}, \mathbf{X}_1, \mathbf{X}_2, \mathbf{D}) \\ = z_{pqt} \sum_{h=1}^{K_2} u_{pqht} \{Y_{pqt} \log(\theta_{pqght}) + (1 - Y_{pqt}) \log(1 - \theta_{pqght})\} \\ + \sum_{m=1}^M s_{tm} \log \Gamma(\alpha_{pgtm} + C_{pgt}) + \text{const.} \end{aligned}$$

Note that $C_{pgt} = C'_{pgt} + z_{pqtg}$ and that, for $x \in \{0, 1\}$, $\Gamma(y + x) = y^x \Gamma(y)$. Since the $z_{pqtg} \in \{0, 1\}$, we can re-express $\log \Gamma(\alpha_{ptmg} + C_{ptg}) = z_{pqtg} \log(\alpha_{ptmg} + C'_{ptg}) + \log \Gamma(\alpha_{ptmg} + C'_{ptg})$ and thus simplify the expression to

$$\begin{aligned} \log P(\mathbf{Y}, \mathbf{L} \mid \mathbf{B}, \boldsymbol{\beta}_1, \boldsymbol{\beta}_2, \boldsymbol{\gamma}, \mathbf{X}_1, \mathbf{X}_2, \mathbf{D}) \\ = z_{pqt} \sum_{h=1}^{K_2} u_{pqht} \{Y_{pqt} \log(\theta_{pqght}) + (1 - Y_{pqt}) \log(1 - \theta_{pqght})\} \\ + z_{pqt} \sum_{m=1}^M s_{tm} \log \Gamma(\alpha_{pgtm} + C'_{pgt}) + \text{const.} \end{aligned}$$

We proceed by taking the expectation of $\tilde{Q}(-z)$ under the variational distribution $\tilde{Q}$:

$$\begin{aligned} \mathbb{E}_{\tilde{Q}} [\log P(\mathbf{Y}, \mathbf{L} \mid \mathbf{B}, \boldsymbol{\beta}_1, \boldsymbol{\beta}_2, \boldsymbol{\gamma}, \mathbf{X}_1, \mathbf{X}_2, \mathbf{D})] \\ = z_{pqt} \sum_{h=1}^{K_2} \mathbb{E}_{\tilde{Q}_2}(u_{pqht}) \{Y_{pqt} \log(\theta_{pqght}) + (1 - Y_{pqt}) \log(1 - \theta_{pqght})\} \\ + z_{pqt} \sum_{m=1}^M \mathbb{E}_{\tilde{Q}_1}(s_{tm}) \log \Gamma(\alpha_{pgtm} + C'_{pgt}) + \text{const.} \end{aligned}$$

The exponential of this expression corresponds to the (unnormalized) parameter vector of a multinomial distribution $\tilde{Q}_2$:

$$\begin{aligned} \hat{\lambda}_{pqtg} &\propto \exp \left[z_{pqgt} \sum_{h=1}^{K_2} \mathbb{E}_{\tilde{Q}_2}(u_{pqht}) \{Y_{pqt} \log(\theta_{pqght}) + (1 - Y_{pqt}) \log(1 - \theta_{pqght})\} \right] \\ &\times \exp \left[z_{pqgt} \sum_{m=1}^M \mathbb{E}_{\tilde{Q}_1}(s_{tm}) \log \Gamma(\alpha_{pgtm} + C'_{pgt}) \right] \end{aligned}$$

Analogously, the update for $\mathbf{u}_{qp}$ is similarly derived:

$$\begin{aligned} &\mathbb{E}_{\tilde{Q}} \log P(\mathbf{Y}, \mathbf{L} \mid \mathbf{B}, \beta_1, \beta_2, \gamma, \mathbf{X}_1, \mathbf{X}_2, \mathbf{D}) \\ &= u_{pqht} \sum_{g=1}^{K_1} \mathbb{E}_{\tilde{Q}_2}(z_{pqgt}) \{Y_{pqt} \log(\theta_{pqght}) + (1 - Y_{pqt}) \log(1 - \theta_{pqght})\} \\ &+ u_{pqht} \sum_{m=1}^M \mathbb{E}_{\tilde{Q}_1}(s_{tm}) \log \Gamma(\alpha_{qht m} + C'_{qht}) + \text{const.} \end{aligned}$$

The exponential of this expression corresponds to the (unnormalized) parameter vector of a multinomial distribution $\tilde{Q}_2$:

$$\begin{aligned} \hat{\delta}_{pqth} &\propto \exp \left[u_{pqht} \sum_{g=1}^{K_1} \mathbb{E}_{\tilde{Q}_2}(z_{pqgt}) \{Y_{pqt} \log(\theta_{pqght}) + (1 - Y_{pqt}) \log(1 - \theta_{pqght})\} \right] \\ &\times \exp \left[u_{pqht} \sum_{m=1}^M \mathbb{E}_{\tilde{Q}_1}(s_{tm}) \log \Gamma(\alpha_{qht m} + C'_{qht}) \right] \end{aligned}$$

3.1.2 E step 2: S

Similar to the last section, collection all terms in eq. (5) that contain s_{tm} for a specific $t > 1$ and m :

$$\begin{aligned} P(\mathbf{Y}, \mathbf{L} \mid \mathbf{B}, \beta, \gamma) &= \Gamma(M\eta + U_m)^{-1} \prod_{m=1}^M \prod_{n=1}^M \Gamma(\eta + U_{mn}) \\ &\times \prod_{p \in V_{1t}} \left[\frac{\Gamma(\xi_{ptm})}{\Gamma(\xi_{ptm} + N_{2t})} \prod_{g=1}^{K_1} \frac{\Gamma(\alpha_{ptgm} + C_{ptg})}{\Gamma(\alpha_{ptgm})} \right]^{s_{tm}} \\ &\times \prod_{q \in V_{2t}} \left[\frac{\Gamma(\xi_{qtm})}{\Gamma(\xi_{qtm} + N_{1t})} \prod_{h=1}^{K_2} \frac{\Gamma(\alpha_{qthm} + C_{qth})}{\Gamma(\alpha_{qthm})} \right]^{s_{tm}} + \text{const.} \end{aligned}$$

Isolating terms that depend on $s_{tm}(n \neq m)$, define

$$\begin{aligned} U'_m &= U_m - s_{tm} \\ U'_{mm} &= U_{mm} - s_{t-1,m}s_{tm} - s_{tm}s_{t+1,m} \\ U'_{nm} &= U_{nm} - s_{t-1,n}s_{tm} \\ U'_{mn} &= U_{mn} - s_{tm}s_{t+1,n} \end{aligned}$$

So that $U'_{ab}, a, b \in \{m, n\}$ counts the number of times the hidden Markov process transitions from a to b except for when the transition happens into or out of time t . Therefore, separating the case where $m = n$ and $m \neq n$, the first two terms on the right hand side can be written as:

$$\begin{aligned} &\Gamma(M\eta + s_{tm} + U'_m)^{-1} \Gamma(\eta + s_{t+1,m}s_{tm} + s_{t-1,m}s_{tm} + U'_{mm}) \\ &\times \prod_{n \neq m}^M \Gamma(\eta + s_{t+1,n}s_{tm} + U'_{mn}) \Gamma(\eta + s_{tm}s_{t-1,n} + U'_{nm}) \end{aligned}$$

Recall that for Gamma function, $\Gamma(y+x) = y^x \Gamma(y)$, for $x \in \{0, 1\}$, therefore this expression becomes

$$(M\eta + U'_m)^{-s_{tm}} \Gamma(M\eta + U'_m)^{-1} \left\{ (\eta + U'_{mm} + 1)^{s_{t+1,m} s_{t-1,m}} (\eta + U'_{mm})^{s_{t-1,m} - s_{t-1,m} s_{t+1,m} + s_{t+1,m}} \right\}^{s_{tm}} \\ \times \Gamma(\eta + U'_{mm}) \prod_{n \neq m}^M (\eta + U'_{mn})^{s_{t+1,n} s_{tm}} \Gamma(\eta + U'_{mn}) \prod_{n \neq m}^M (\eta + U'_{nm})^{s_{tm} s_{t-1,n}} \Gamma(\eta + U'_{nm})$$

To see why

$$\Gamma(\eta + s_{t+1,m} s_{tm} + s_{t-1,m} s_{tm} + U'_{mm}) = \\ \left\{ (\eta + U'_{mm} + 1)^{s_{t+1,m} s_{t-1,m}} (\eta + U'_{mm})^{s_{t-1,m} - s_{t-1,m} s_{t+1,m} + s_{t+1,m}} \right\}^{s_{tm}} \Gamma(\eta + U'_{mm}) \quad (*)$$

Recall that $s_{tm} = \mathbb{I}(s_t = m) \in \{0, 1\}$, and consider in turn the following cases:

- 1) When $s_{tm} = 0$, equation (*) simplifies to $\Gamma(\eta + U'_{mm}) = 1 \times \Gamma(\eta + U'_{mm})$, for any values of $s_{t-1,m}$ and $s_{t+1,m}$
- 2) When $s_{tm} = 1$,
 - i) $s_{t-1,m} = 0$ and $s_{t+1,m} = 0$, equation (*) simplifies to: $\Gamma(\eta + U'_{mm}) = (1 \times 1)^1 \Gamma(\eta + U'_{mm})$
 - ii) $s_{t-1,m} = 1$ and $s_{t+1,m} = 0$, equation (*) becomes (recall the recursive property of the Gamma function):
$$\Gamma(\eta + U'_{mm} + 1) = (1 \times (\eta + U'_{mm} + 1))^1 \Gamma(\eta + U'_{mm})$$
 - iii) $s_{t-1,m} = 1$ and $s_{t+1,m} = 0$. This is analogous to ii)
 - iv) $s_{t-1,m} = 1$ and $s_{t+1,m} = 1$, equation (*) becomes (applying the recursive property twice):

$$\Gamma(\eta + U'_{mm} + 1 + 1) = (\eta + U'_{mm} + 1) \Gamma(\eta + U'_{mm} + 1) \\ = (\eta + U'_{mm} + 1)(\eta + U'_{mm}) \Gamma(\eta + U'_{mm}) \\ = [(\eta + U'_{mm} + 1)^1 (\eta + U'_{mm})^1]^1 \Gamma(\eta + U'_{mm})$$

Again focus on the terms that are specific to a t and m ,

$$P(\mathbf{Y}, \mathbf{L} \mid \mathbf{B}, \boldsymbol{\beta}, \boldsymbol{\gamma}) \\ = (M\eta + U'_m)^{-s_{tm}} \left\{ (\eta + U'_{mm} + 1)^{s_{t+1,m} s_{t-1,m}} (\eta + U'_{mm})^{s_{t-1,m} - s_{t-1,m} s_{t+1,m} + s_{t+1,m}} \right\}^{s_{tm}} \\ \times \prod_{n \neq m}^M (\eta + U'_{mn})^{s_{t+1,n} s_{tm}} (\eta + U'_{nm})^{s_{tm} s_{t-1,n}} \\ \times \prod_{p \in V_{1t}} \left[\frac{\Gamma(\xi_{ptm})}{\Gamma(\xi_{ptm} + N_{2t})} \prod_{g=1}^{K_1} \frac{\Gamma(\alpha_{ptgm} + C_{ptg})}{\Gamma(\alpha_{ptgm})} \right]^{s_{tm}} \\ \times \prod_{q \in V_{2t}} \left[\frac{\Gamma(\xi_{qtm})}{\Gamma(\xi_{qtm} + N_{1t})} \prod_{h=1}^{K_2} \frac{\Gamma(\alpha_{qthm} + C_{qth})}{\Gamma(\alpha_{qthm})} \right]^{s_{tm}} + \text{const.}$$

Taking log and expectation under $\tilde{Q}$ w.r.t. variables do not contain s_{tm} :

$$\begin{aligned}
\log \hat{\phi}_{tm} = & -s_{tm} \mathbb{E}_{\tilde{Q}_1} [\log (M\eta + U'_m)] + s_{tm} \phi_{t+1,m} \phi_{t-1,m} \mathbb{E}_{\tilde{Q}_1} [\log (\eta + U'_{mm} + 1)] \\
& + s_{tm} (\phi_{t-1,m} - \phi_{t-1,m} \phi_{t+1,m} + \phi_{t+1,m}) \mathbb{E}_{\tilde{Q}_1} [\log (\eta + U'_{mm})] \\
& + s_{tm} \sum_{n \neq m}^M \phi_{t+1,n} \mathbb{E}_{\tilde{Q}_1} [\log (\eta + U'_{mn})] + s_{tm} \sum_{n \neq m}^M \phi_{t-1,n} \mathbb{E}_{\tilde{Q}_1} [\log (\eta + U'_{nm})] \\
& + s_{tm} \sum_{p \in V_{1t}} \left[\frac{\Gamma(\xi_{ptm})}{\Gamma(\xi_{ptm} + N_{2t})} \right] + s_{tm} \sum_{p \in V_{t1}} \sum_{g=1}^{K_1} \mathbb{E}_{\tilde{Q}_2} \left[\log \left[\frac{\Gamma(\alpha_{ptmg} + C_{ptg})}{\Gamma(\alpha_{ptmg})} \right] \right] \\
& + s_{tm} \sum_{q \in V_{2t}} \left[\frac{\Gamma(\xi_{qtm})}{\Gamma(\xi_{qtm} + N_{1t})} \right] + s_{tm} \sum_{q \in V_{t2}} \sum_{h=1}^{K_2} \mathbb{E}_{\tilde{Q}_2} \left[\log \left[\frac{\Gamma(\alpha_{qtmh} + C_{qth})}{\Gamma(\alpha_{qtmh})} \right] \right] + \text{const}
\end{aligned}$$

So that the m th element of the parameter vector for $\tilde{Q}_1(s_t | \phi_{tm})$ is (so we could treat the expectations w.r.t. to $\tilde{Q}_2$ as constant):

$$\begin{aligned}
\hat{\phi}_{tm} \propto & \exp \left[-\mathbb{E}_{\tilde{Q}_1} [\log (M\eta + U'_m)] \right] \exp \left[\phi_{t+1,m} \phi_{t-1,m} \mathbb{E}_{\tilde{Q}_1} [\log (\eta + U'_{mm} + 1)] \right] \\
& \times \exp \left[(\phi_{t-1,m} - \phi_{t-1,m} \phi_{t+1,m} + \phi_{t+1,m}) \mathbb{E}_{\tilde{Q}_1} [\log (\eta + U'_{mm})] \right] \\
& \times \prod_{n \neq m} \exp \left[\phi_{t+1,n} \mathbb{E}_{\tilde{Q}_1} [\log (\eta + U'_{mn})] \right] \exp \left[\phi_{t-1,n} \mathbb{E}_{\tilde{Q}_1} [\log (\eta + U'_{nm})] \right] \\
& \times \prod_{p \in V_{1t}} \left[\frac{\Gamma(\xi_{ptm})}{\Gamma(\xi_{ptm} + N_{2t})} \prod_{g=1}^{K_1} \frac{\Gamma(\alpha_{ptgm} + C_{ptg})}{\Gamma(\alpha_{ptgm})} \right] \\
& \times \prod_{q \in V_{2t}} \left[\frac{\Gamma(\xi_{qtm})}{\Gamma(\xi_{qtm} + N_{1t})} \prod_{h=1}^{K_2} \frac{\Gamma(\alpha_{qthm} + C_{qth})}{\Gamma(\alpha_{qthm})} \right]
\end{aligned}$$

3.2 The M-Steps

3.2.1 The Lower Bound

The full expression of the lower bound can be written as:

$$\begin{aligned}
\mathcal{L}(\tilde{Q}) &= \mathbb{E}_{\tilde{Q}} [\log P(\mathbf{Y}, \mathbf{L} \mid \beta_1, \beta_2, \gamma, \mathbf{B}, \mathbf{X}_1, \mathbf{X}_2, \mathbf{D})] - \mathbb{E}_{\tilde{Q}} [\log \tilde{Q}(\mathbf{L} \mid \Phi, \Lambda, \Delta)] \\
&= \log(P(s_1)) + \log \Gamma(M\eta) - \sum_m \mathbb{E}_{\tilde{Q}} [\log \Gamma(M\eta + U_m)] + \sum_{m,n} \mathbb{E}_{\tilde{Q}} [\log \Gamma(\eta + U_{m,n})] - \log \Gamma(\eta) \\
&+ \sum_{t,m} \phi_{tm} \sum_{p \in V_{1,t}} [\Gamma(\xi_{ptm}) - \Gamma(\xi_{ptm} + N_{2t})] + \sum_{t,m} \phi_{tm} \sum_{p \in V_{1,t}} \sum_{g=1}^{K_1} [\mathbb{E}_{\tilde{Q}} [\log \Gamma(\alpha_{ptgm} + C_{ptg})] - \log \Gamma(\alpha_{ptgm})] \\
&+ \sum_{t,m} \phi_{tm} \sum_{q \in V_{2,t}} [\Gamma(\xi_{qtm}) - \Gamma(\xi_{qtm} + N_{1t})] + \sum_{t,m} \phi_{tm} \sum_{q \in V_{2,t}} \sum_{h=1}^{K_2} [\mathbb{E}_{\tilde{Q}} [\log \Gamma(\alpha_{qtgm} + C_{qth})] - \log \Gamma(\alpha_{qtgm})] \\
&+ \sum_t \sum_{(p,q) \in V_{1t} \times V_{2t}} \sum_{g=1}^{K_1} \sum_{h=1}^{K_2} \lambda_{pq,g} \delta_{pq,h} \{y_{pqt} \log \theta_{pqgh} + (1 - y_{pqt}) \log (1 - \theta_{pqgh})\} \\
&- \sum_{g=1}^{K_1} \sum_{h=1}^{K_2} \frac{(B_{gh} - \mu_{gh})^2}{2\sigma_{gh}^2} - \sum_{j=1}^{J_d} \frac{(\gamma_j - \mu_\gamma)^2}{2\sigma_\gamma^2} - \sum_{g=1}^{K_1} \sum_{j=1}^{J_{1x}} \sum_{m=1}^M \frac{(\beta_{1gjm} - \mu_{\beta_1})^2}{2\sigma_{\beta_1}^2} - \sum_{h=1}^{K_2} \sum_{j=1}^{J_{2x}} \sum_{m=1}^M \frac{(\beta_{2hjm} - \mu_{\beta_2})^2}{2\sigma_{\beta_2}^2} \\
&- \sum_{g=1}^{K_1} \sum_{m=1}^M \frac{\nu+1}{2} \log(1 + \frac{\beta_{1.0}^2}{\nu}) - \sum_{h=1}^{K_2} \sum_{m=1}^M \frac{\nu+1}{2} \log(1 + \frac{\beta_{2.0}^2}{\nu}) \\
&- \sum_{t,m} \phi_{tm} \log(\phi_{t,m}) - \sum_{(p,q) \in V_{1t} \times V_{2t}} \sum_{g=1}^{K_1} \sum_{h=1}^{K_2} \{\lambda_{pqg} \log(\lambda_{pqg}) - \delta_{pqh} \log(\delta_{pqh})\}
\end{aligned} \tag{8}$$

3.2.2 M step 1: B

Collect the terms that contain B_{gh} in the lower bound:

$$\begin{aligned}
\mathcal{L}(\tilde{Q}) &= \sum_{t=1}^T \sum_{p,q \in V_{1t} \times V_{2t}} \sum_{g,h=1}^K \lambda_{pqtg} \delta_{qpth} \{y_{pqt} \log \theta_{pqtgh} + (1 - y_{pqt}) \log (1 - \theta_{pqtgh})\} \\
&- \sum_{g=1}^{K_1} \sum_{h=1}^{K_2} \frac{(B_{gh} - \mu_{gh})^2}{2\sigma_{gh}^2} + \text{const.}
\end{aligned}$$

We optimize this lower bound with respect to $\mathbf{B}_{gh}$ using a gradient-based numerical optimization method. The corresponding gradient is given by,

$$\frac{\partial \mathcal{L}_{B_{gh}}}{\partial B_{gh}} = \sum_{t=1}^T \sum_{p,q \in V_{1t} \times V_{2t}} \lambda_{pqtg} \delta_{qpth} (y_{pqt} - \theta_{pqtgh}) - \frac{B_{gh} - \mu_{B_{gh}}}{\sigma_{B_{gh}}^2}$$

3.2.3 M step 2: γ

Collect the terms that contain γ in the lower bound (note that $\theta_{pqtgh} = \text{logit}^{-1}(B_{z_{pqt}, u_{pqt}} + \mathbf{d}_{pq}^\top \gamma)$ is also a function of γ), J_d is the number of dyadic covariates:

$$\begin{aligned}
\mathcal{L}(\tilde{Q}) &= \sum_{t=1}^T \sum_{p,q \in V_{1t} \times V_{2t}} \sum_{g=1}^{K_1} \sum_{h=1}^{K_2} \lambda_{pqtg} \delta_{qpth} \{y_{pqt} \log \theta_{pqtgh} + (1 - y_{pqt}) \log (1 - \theta_{pqtgh})\} \\
&- \sum_j \frac{(\gamma_j - \mu_\gamma)^2}{2\sigma_\gamma^2} + \text{const.}
\end{aligned}$$

Similarly, we use a numerical optimization algorithm based on the following gradient to optimize this expression with respect to γ_j (the j th element of the γ vector). The corresponding gradient is given by,

$$\frac{\partial \mathcal{L}_{\gamma_j}}{\partial \gamma_j} = \sum_{t=1}^T \sum_{p,q \in V_{1t} \times V_{2t}} \sum_{g=1}^{K_1} \sum_{h=1}^{K_2} \lambda_{pqtg} \delta_{qpth} \mathbf{d}_{pqtj}^\top (y_{pqt} - \theta_{pqtgh}) - \frac{\gamma_j - \mu_\gamma}{\sigma_\gamma^2}$$

3.2.4 M step 3: β_{1m} and β_{2m}

First, collect all terms that contain β_{1gm} and roll the rest of the terms into a constant. J_{1x} is the number of monadic covariates for family 1 (excluding the intercept term), and J_{2x} is the number of monadic covariates for family 2:

$$\begin{aligned} \mathcal{L}(\tilde{Q}) = & \sum_{t=1}^T \sum_{m=1}^M \phi_{tm} \sum_{p \in V_{1t}} [\log \Gamma(\xi_{ptm}) - \log \Gamma(\xi_{ptm} + N_{2t})] \\ & + \sum_{t,m} \phi_{tm} \sum_{p \in V_{1t}} \sum_{g=1}^{K_1} \left[\mathbb{E}_{\tilde{Q}_2} [\log \Gamma(\alpha_{ptgm} + C_{ptg})] - \log \Gamma(\alpha_{ptgm}) \right] \\ & - \sum_{g=1}^{K_1} \sum_{j=1}^{J_{1x}} \sum_{m=1}^M \frac{(\beta_{1gjm} - \mu_{\beta_1})^2}{2\sigma_{\beta_1}^2} - \sum_{g=1}^{K_1} \sum_{m=1}^M \frac{\nu+1}{2} \log(1 + \frac{\beta_{1.0}^2}{\nu}) + \text{const.} \end{aligned}$$

ν is the degree of freedom of the t distribution the intercept term $\beta_{1.0}$ is drawn from. Following Gelman et al. (2008), we set the default value of $\nu = 7$. No closed-form solution exists for an optimum with respect to β_{1mgj} , but a gradient-based algorithm can be implemented to maximize the above expression. The corresponding gradient with respect to each element in vector β_{1gm} is given by:

$$\begin{aligned} \frac{\partial \mathcal{L}(\tilde{Q})}{\partial \beta_{1mgj}} = & \sum_{t=1}^T \phi_{tm} \sum_{p \in V_{1t}} \alpha_{ptmg} x_{1ptj} \left(\mathbb{E}_{\tilde{Q}_2} [\psi(\alpha_{ptmg} + C_{ptg}) - \psi(\alpha_{ptgm})] \right. \\ & \left. + [\psi(\xi_{ptm}) - \psi(\xi_{ptm} + N_{2t})] \right) - \frac{\beta_{1mgj} - \mu_{\beta_1}}{\sigma_{\beta_1}^2} - \sum_{g=1}^{K_1} \sum_{m=1}^M \frac{(1+\nu)\beta_{1.0}}{\nu + \beta_{1.0}^2} \end{aligned}$$

Here, ψ is the digamma function. Similarly for β_{2hm} , collect all the relevant terms yield:

$$\begin{aligned} \mathcal{L}(\tilde{Q}) = & \sum_{t=1}^T \sum_{m=1}^M \phi_{tm} \sum_{q \in V_{2t}} [\log \Gamma(\xi_{qtm}) - \log \Gamma(\xi_{qtm} + N_{1t})] \\ & + \sum_{q \in V_{2t}} \sum_{h=1}^{K_2} \left[\mathbb{E}_{\tilde{Q}_2} [\log \Gamma(\alpha_{qhtm} + C_{qth})] - \log \Gamma(\alpha_{qhtm}) \right] \\ & - \sum_{h=1}^{K_2} \sum_{j=1}^{J_{2x}} \sum_{m=1}^M \frac{(\beta_{2hjm} - \mu_{\beta_2})^2}{2\sigma_{\beta_2}^2} - \sum_{h=1}^{K_2} \sum_{m=1}^M \frac{\nu+1}{2} \log(1 + \frac{\beta_{2.0}^2}{\nu}) + \text{const.} \end{aligned}$$

With the corresponding gradient:

$$\begin{aligned} \frac{\partial \mathcal{L}(\tilde{Q})}{\partial \beta_{2mhj}} = & \sum_{t=1}^T \phi_{tm} \sum_{q \in V_{2t}} \alpha_{qtmh} x_{2ptj} \left(\mathbb{E}_{\tilde{Q}_2} [\psi(\alpha_{qtmh} + C_{qth}) - \psi(\alpha_{qthm})] \right. \\ & \left. + [\psi(\xi_{qtm}) - \psi(\xi_{qtm} + N_{1t})] \right) - \frac{\beta_{2mhj} - \mu_{\beta_2}}{\sigma_{\beta_2}^2} - \sum_{h=1}^{K_2} \sum_{m=1}^M \frac{(1+\nu)\beta_{2.0}}{\nu + \beta_{2.0}^2} \end{aligned}$$

3.3 Standard Error Computation

3.3.1 Hessian for γ

Restricted to terms that involve γ , we have shown that

$$\frac{\partial \mathcal{L}(\tilde{Q})}{\partial \gamma_j} = \sum_{t=1}^T \sum_{p,q \in V_{1t} \times V_{2t}} \sum_{g=1}^{K_1} \sum_{h=1}^{K_2} \lambda_{pqtg} \delta_{qpth} \mathbf{d}_{pqtj}^\top (y_{pqt} - \theta_{pqtgh}) - \frac{\gamma_j - \mu_\gamma}{\sigma_\gamma^2}$$

Then,

$$\frac{\partial^2 \mathcal{L}(\tilde{Q})}{\partial \gamma_j \partial \gamma_{j'}} = \sum_{t=1}^T \sum_{p,q \in V_{1t} \times V_{2t}} \sum_{g=1}^{K_1} \sum_{h=1}^{K_2} -\mathbf{d}_{pqtj}^\top \mathbf{d}_{pqtj'} [\bar{\theta}_{pqtgh}(1 - \bar{\theta}_{pqtgh})] - \sigma_\gamma^{-2} \delta_{jj'}$$

Here, $\delta_{jj'}$ is the Kronecker delta function, and

$$\bar{\theta}_{pqtgh} = \mathbb{E}_{\tilde{Q}} [\theta_{pqtgh}] = \hat{\lambda}_{pqtg}^\top \hat{\mathbf{B}} \hat{\delta}_{qpth} + \mathbf{d}_{pqt}^\top \gamma$$

is a closed-form solution to the expectation over $\tilde{Q}$.

3.3.2 Hessian for β_1 and β_2

First, we focus on family 1 coefficients, which is β_1 . For coefficients in the same group g :

$$\begin{aligned} \frac{\partial \mathcal{L}(\tilde{Q})}{\partial \beta_{1mgj}} &= \sum_{t=1}^T \phi_{tm} \sum_{p \in V_{1t}} \alpha_{ptmg} x_{1ptj} \left(\mathbb{E}_{\tilde{Q}_2} \left[\check{\psi}(\alpha_{ptmg} + C_{ptg}) - \check{\psi}(\alpha_{ptgm}) \right] \right. \\ &\quad \left. + \left[\check{\psi}(\xi_{ptm}) - \check{\psi}(\xi_{ptm} + N_{2t}) \right] \right) - \frac{\beta_{1mgj} - \mu_{\beta_1}}{\sigma_{\beta_1}^2} \end{aligned}$$

So,

$$\begin{aligned} \frac{\partial^2 \mathcal{L}(\tilde{Q})}{\partial \beta_{1mgj} \partial \beta_{1mgj'}} &= \sum_{t=1}^T \phi_{tm} \sum_{p \in V_{1t}} x_{1ptj} x_{1ptj'} \alpha_{ptmg} \left\{ \mathbb{E}_{\tilde{Q}_2} \left[\check{\psi}(\alpha_{ptmg} + C_{ptg}) - \check{\psi}(\alpha_{ptgm}) \right] \right. \\ &\quad \left. + \check{\psi}(\xi_{ptm}) - \check{\psi}(\xi_{ptm} + N_{2t}) \right. \\ &\quad \left. + \alpha_{ptmg} \left[\mathbb{E}_{\tilde{Q}_2} \left[\check{\psi}_1(\alpha_{ptmg} + C_{ptg}) - \check{\psi}_1(\alpha_{ptgm}) \right] + \check{\psi}_1(\xi_{ptm}) - \check{\psi}_1(\xi_{ptm} + N_{2t}) \right] \right\} \end{aligned}$$

Here, $\check{\psi}_1$ is the trigamma function. For coefficients in different latent groups g and g' ,

$$\frac{\partial^2 \mathcal{L}(\tilde{Q})}{\partial \beta_{1mgj} \partial \beta_{1mg'j'}} = \sum_{t=1}^T \phi_{tm} \sum_{p \in V_{1t}} x_{1ptj} x_{1ptj'} \alpha_{ptmg} \alpha_{ptmg'} \left(\check{\psi}_1(\xi_{ptm}) - \check{\psi}_1(\xi_{ptm} + N_{2t}) \right)$$

The Hessian for β_2 can be derived similarly.

Unlike γ , there are no closed-form solutions for the expectations involved in the Hessian for β_1 and β_2 . To approximate them, we take S samples from the Poisson-Binomial distribution of C_{ptg} , and we get $C_{ptg}^{(s)}$ ($s \in 1 \dots S$), and let

$$\begin{aligned} \mathbb{E}_{\tilde{Q}_2} \left[\check{\psi}(\alpha_{ptmg} + C_{ptg}) \right] &\approx \frac{1}{S} \sum_S \left(\check{\psi}(\alpha_{ptmg} + C_{ptg}^{(s)}) \right) \\ \mathbb{E}_{\tilde{Q}_2} \left[\check{\psi}_1(\alpha_{ptmg} + C_{ptg}) \right] &\approx \frac{1}{S} \sum_S \left(\check{\psi}_1(\alpha_{ptmg} + C_{ptg}^{(s)}) \right) \end{aligned}$$